

MediRAG – Multimodal Medical Retrieval Assistant

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Title:- (SIH25231) Design and build a multimodal Retrieval-Augmented Generation (RAG) system leveraging a Large Language Model (LLM) for OFFLINE mode that can ingest, index, and query diverse data formats such as DOC, PDF, Images and voice recordings within a unified semantic retrieval framework.

Brief Description of Project: -

Project Overview

Healthcare professionals and students often require fast access to accurate information from multiple medical resources such as **PDF reports, clinical guidelines, medical notes (DOC), images, and voice recordings (doctor-patient conversations / dictation)**. However, most systems depend heavily on the internet and cloud-based search engines, which is not suitable in many cases like **remote hospitals, restricted environments, defense/secure medical labs, and offline emergency situations**.

To solve this, our project **MediRAG – Multimodal Medical Retrieval Assistant** proposes an **multimodal Retrieval-Augmented Generation (RAG) system** that can ingest, index, and answer queries from multiple types of medical data. The system will use a **Large Language Model (LLM) running in OFFLINE mode** along with semantic retrieval methods to provide accurate, evidence-based answers with references from stored documents.

This project aligns with the SIH problem statement of developing a **unified semantic retrieval framework** capable of handling multiple data types and enabling effective querying in offline mode.

Problem Statement

Medical knowledge is distributed across multiple sources and formats. Traditional search mechanisms:

- work mainly on keyword matching,
- cannot understand medical context properly,
- fail to connect information across documents,
- and typically require online access.

Also, medical practitioners may need to:

- search within hundreds of PDFs quickly,
- retrieve context from patient files,
- compare symptoms across reports,
- and interpret supporting medical images.

Hence, there is a strong need for an offline medical intelligent assistant which can:

- store and process multimodal medical data,
- understand natural language questions,
- retrieve most relevant context,
- and generate helpful, medically meaningful responses.

Proposed Solution

We propose an **Offline Multimodal RAG System** customized for the **Medical domain**. The complete system will work in 3 main layers:

1. Data Ingestion Layer

This module will allow uploading and handling multiple data formats:

- **PDF files:** clinical guidelines, prescriptions, research papers, medical manuals
- **DOC/DOCX files:** doctor notes, reports, summaries
- **Images:** lab report images
- **Voice recordings:** doctor dictation, medical lectures, recorded patient symptoms

Each uploaded file will be processed as follows:

- Text extracted from PDFs/DOCs using document parsing tools
- Images converted into embeddings using a vision model
- Audio converted to text using offline speech-to-text (ASR) models
- Metadata captured such as document type, title, upload time, category, and patient/department tags

This helps in building a structured knowledge base from unstructured medical data.

2. Indexing & Semantic Retrieval Layer (Offline Vector Store)

After preprocessing, the extracted content will be converted into **vector embeddings** (numerical representation of meaning).

We will build an **offline vector database** using tools like:

- FAISS / ChromaDB (offline supported)

Indexing will support:

- **Medical semantic similarity search**
- **Multi-format retrieval**

- **Context chunking** (splitting long medical documents into meaningful sections)
- **Tag and filter-based retrieval** (example: only cardiology documents)

The retrieval system will ensure that when a user asks a query, it fetches the **top relevant text passages/images/audio-derived content** from the local database.

3. Generation Layer (LLM + RAG Pipeline)

The retrieved information will be passed into an **offline LLM** to generate a final answer.

This stage ensures:

- answers are based on **retrieved evidence**
- reduces hallucinations
- provides **source references**
- gives structured output such as:
 - symptoms
 - possible causes
 - recommended steps
 - preventive measures
 - medicine information (general)
 - emergency warning signs

Example queries supported:

- “What are the symptoms and treatment for dengue?”
- “Explain this lab report and what it indicates.”

4.Flow Diagram

